HOMEWORK #4

Fall 2025
EIN 4360: Facilities Planning & Material Handling

Instructor: **Nicolas Grimaldi**
Due: **Wednesday, Mar 11th, at 11:59 PM**

Lecture 7: Material Handling

Problem 1. Milburn Pennybags is considering setting up a new money printing subsidiary for the US Department of Treasury. Specifically, Mr. Pennybags will need to print out coins and cash and store them within his facility. He is very particular about transporting the cash from the manufacturing line to the storage and wants to make sure that each bill arrives in perfect condition. However, he does not really care about coins. Coins and cash have different manufacturing and storage locations. For cash, the movement from manufacturing to storage is not too long and always follows a fixed route. For coins, the storage location often varies between a couple of locations, which are not confined to one area. Because Mr. Pennybags has a lot of starting capital, he would rather make a large initial investment and see returns in the future.

a) What type of material handling equipment should Mr. Pennybags use to **move** the cash to storage? What about the coins?

b) What type of storage equipment should Mr. Pennybags use to **store** the cash? What about the coins?

Be specific in your answer, if they are industrial trucks, tell me which and why those are preferred over others. Only consider the material handling equipment we discussed in class (those in the slide images).

Lecture 8: Warehouse Operations & Layout

Problem 2. Consider the facility below (Fig. 2.1), which has 32 bays. Each day is 25'x25'. Four product types (A, B, C, and D) need to be stored. The facility uses dedicated storage where each bay is only assigned to one product type. Assume material comes in through dock 1 and is shipped through docks 2 & 3, where dock 2 is used twice as often as dock 3. The area requirements and weekly load rates are shown in the table below (Table 2.1). Determine the layout that will minimize the average distance traveled per week for this facility.

Product	Area (ft^2)	Weekly Load Rate
A	4,375	500
B	7,500	450
C	1,875	300
D	6,250	600

Table 2.1. Products' area requirements and load rates.

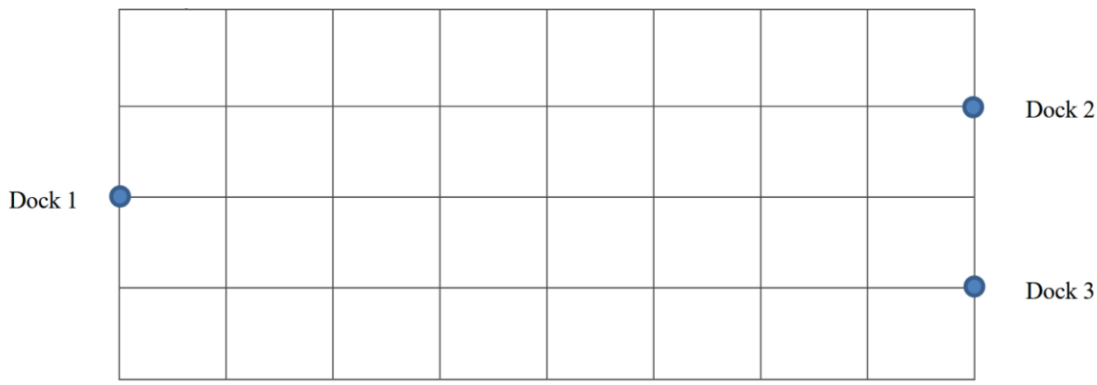

Fig. 2.1. Facility's storage layout with docks, where one grid corresponds to one bay.

Problem 3. Answer the following questions about the dimensions of the ASRS. *Hint: note the 0.5' gap between the rows.*

- **a)** What is the total cube of the facility?
- **b)** What is the aisle cube?
- **c)** What is the percentage of total volume allocated to the aisles?

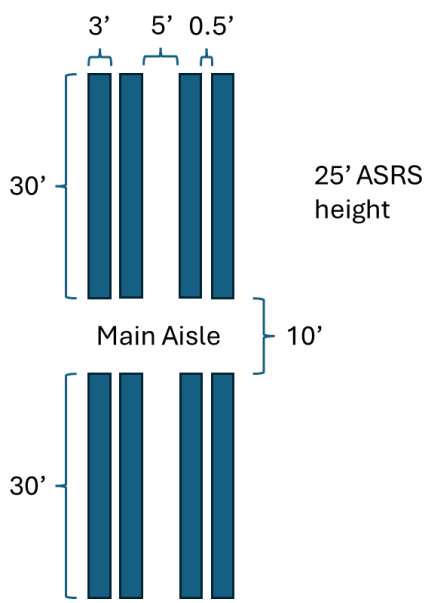

Figure 3.1. Finished ASRS dimensions.

Lecture 9: ASRS

Problem 4. You want to store a unit on a pallet of 24”x20”x12” (L x W x H). You plan to build an ASRS with exactly 10 cranes and want to store up to 100,000 pallets. Clearances are 3’ from the ceiling, 1’ from the floor, and 0.5’ between pallets. Six and seven are your favorite numbers, so you want to have exactly 67 pallets stacked in each bay.

Find the **height, width, and length** of the AS/R system.

Hint: Unlike the previous problem, racks are only on one side of the building (like the problems we did in class). Also, make sure you use the clearance values in your calculations.